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Experiment 2

Aim: To containerise the application using docker

This experiment maps to LO43
LO3:Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Theory

Docker

Docker is an open-source containerization platform used to develop, package, and deploy applications in isolated environments called **containers**. It allows developers to bundle an application along with its libraries, dependencies, and configuration files into a single package, ensuring that the application runs consistently across different systems. Since Docker containers share the host operating system's kernel, they are lightweight and start much faster than traditional virtual machines. Docker simplifies application deployment, improves portability, and supports modern DevOps practices such as Continuous Integration and Continuous Deployment (CI/CD). It is widely used in cloud computing, microservices, and scalable application development.

Installation Steps

1. Download **Docker Desktop** from the official Docker website.
2. Run the installer and select **Use WSL 2 instead of Hyper-V** (recommended).
3. Complete the installation and restart your computer if prompted.
4. Launch **Docker Desktop** and wait until it displays **Docker Desktop is running**.
5. Open **Command Prompt**, **PowerShell**, or **Windows Terminal**.
6. Verify the installation using:
   docker --version
7. Check that Docker is running correctly:
   docker run hello-world
   If the "Hello from Docker!" message appears, Docker has been installed successfully.

Containerization

Containerization is the process of packaging an application together with all its required dependencies into a lightweight, isolated unit called a **container**. Each container runs independently while sharing the host operating system's kernel, making it more efficient than traditional virtual machines. Because containers do not require a separate operating system, they consume fewer resources and launch quickly. Containerization ensures that applications behave consistently across development, testing, and production environments, improving portability, scalability, and reliability. Docker is one of the most widely used platforms for containerization, while Kubernetes is commonly used to manage and orchestrate multiple containers.

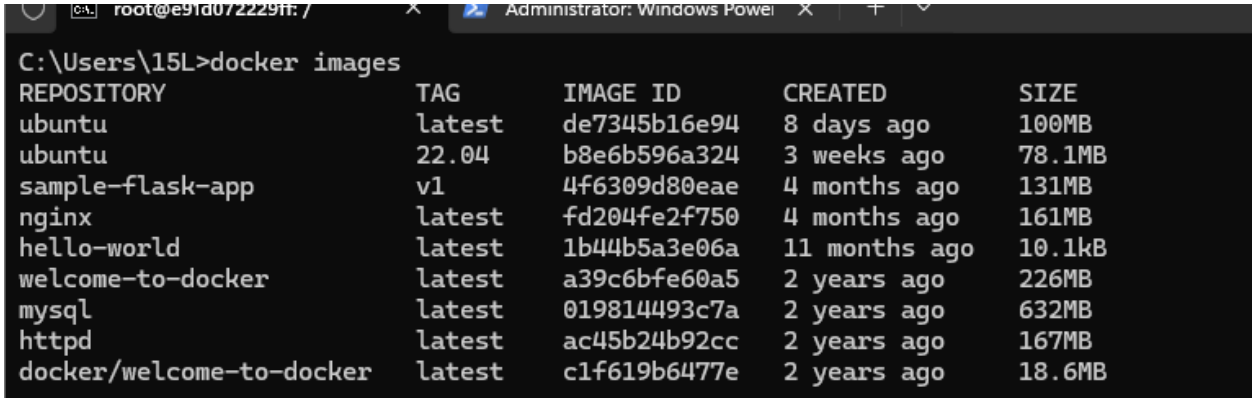
Difference Between Docker Image and Docker Container

Docker Image	Docker Container
A Docker image is a read-only template used to create containers.	A Docker container is a running instance of a Docker image.
It contains the application code, libraries, dependencies, and configuration files.	It executes the application in an isolated environment.
Images are created using a **Dockerfile**.	Containers are created from Docker images.
An image is static and cannot be modified while running.	A container is dynamic and changes during execution.
One image can be used to create multiple containers.	Multiple containers can run simultaneously from the same image.
Images are stored in repositories such as Docker Hub.	Containers run on the host system using the Docker Engine.
Example: `ubuntu`, `nginx`, `python`.	Example: A running Ubuntu shell or an Nginx web server container.

Commands

1. Version : Checks the version of docker installed

2. Docker search ubuntu - - filter stars= 500 : Gives the images that are equal to that number or more



3. Docker pull ubuntu: Downloads images from docker



4. Docker run ubuntu /bin/bash: creates a new Ubuntu container and opens an interactive Bash terminal inside it
5. Docker ps- Displays a list of all currently running Docker containers.

```
C:\Users\15L>docker images
REPOSITORY    TAG       IMAGE ID       CREATED         SIZE
ubuntu        latest    de7345b16e94   6 days ago      100MB
nginx         latest    fd204fe2f750   4 months ago    161MB
hello-world   latest    1b44b5a3e06a   11 months ago   10.1kB
ubuntu        <none>    ca2b0f26964c   2 years ago     77.9MB

C:\Users\15L>docker run ubuntu/bin/bash
Unable to find image 'ubuntu/bin/bash:latest' locally
docker: Error response from daemon: pull access denied for ubuntu/bin/bash, repository does not exist or may require 'do
cker login': denied: requested access to the resource is denied.
See 'docker run --help'.

C:\Users\15L>docker run ubuntu /bin/bash

C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\15L>docker run -it
"docker run" requires at least 1 argument.
See 'docker run --help'.

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Create and run a new container from an image

C:\Users\15L>docker run -it ubuntu /bin/bash
root@d5e54f781411:/#
```

6. Docker run: Create and start a new container from a Docker image.

```
C:\Users\15L>docker run --name mycontainer -it ubuntu bash
root@f1d175be2472:/# exit
exit
```

7. Docker stop: Sends a termination signal to the container, allowing the application inside the container to shut down properly before the container stops.

```
C:\Users\15L>docker stop mycontainer
mycontainer
```

8. Docker start:Used to restart a stopped container.

```
C:\Users\15L>docker start mycontainer
mycontainer
```

9. Docker inspect :Used to display detailed information about a Docker object, such as a container, image, network, or volume.

```
C:\Users\15L>docker inspect mycontainer
[
    {
        "Id": "f1d175be24729df20eb456b47db1df7ad5866b3226a413187f8b88f18917e0c5",
        "Created": "2026-07-20T10:11:57.168507052Z",
        "Path": "bash",
        "Args": [],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 1869,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-20T10:14:07.279713661Z",
            "FinishedAt": "2026-07-20T10:13:51.259521275Z"
        },
        "Image": "sha256:de7345b16e942e22044c6ba053020ec85ae879984860a9918517d54eb6cef851",
        "ResolvConfPath": "/var/lib/docker/containers/f1d175be24729df20eb456b47db1df7ad5866b3226a413187f8b88f18917e0c5/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/f1d175be24729df20eb456b47db1df7ad5866b3226a413187f8b88f18917e0c5/hostname",
        "HostsPath": "/var/lib/docker/containers/f1d175be24729df20eb456b47db1df7ad5866b3226a413187f8b88f18917e0c5/hosts",
        "LogPath": "/var/lib/docker/containers/f1d175be24729df20eb456b47db1df7ad5866b3226a413187f8b88f18917e0c5/f1d175be24729df20eb456b47db1df7ad5866b3226a413187f8b88f18917e0c5-json.log",
        "Name": "/mycontainer",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "default",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
```

```
            "WorkingDir": "",
            "Entrypoint": null,
            "OnBuild": null,
            "Labels": {
                "org.opencontainers.image.created": "2026-07-13T16:06:30.499069+00:00",
                "org.opencontainers.image.description": "The Ubuntu container image maintained by Canonical\n\nUbuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all yo
ur internet connected things.\nIt is the world's most popular operating system across public clouds and OpenStack clouds.\nIt is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu
 can run your containers at scale.\nFast, secure and simple, Ubuntu powers millions of PCs worldwide.\n",
                "org.opencontainers.image.title": "ubuntu",
                "org.opencontainers.image.version": "26.04"
            }
        },
        "NetworkSettings": {
            "Bridge": "",
            "SandboxID": "8b0dd6144bb90f7e585f2b66fd3f11adbcf94bc74e50d85585f283c8bd4d6a8e",
            "SandboxKey": "/var/run/docker/netns/8b0dd6144bb9",
            "Ports": {},
            "HairpinMode": false,
            "LinkLocalIPv6Address": "",
            "LinkLocalIPv6PrefixLen": 0,
            "SecondaryIPAddresses": null,
            "SecondaryIPv6Addresses": null,
            "EndpointID": "730350d5da4b961daefb500c7803e4a37565a1a56cb7408a570fea5e34f822aa6",
            "Gateway": "172.17.0.1",
            "GlobalIPv6Address": "",
            "GlobalIPv6PrefixLen": 0,
            "IPAddress": "172.17.0.2",
            "IPPrefixLen": 16,
            "IPv6Gateway": "",
            "MacAddress": "02:42:ac:11:00:02",
            "Networks": {
                "bridge": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": null,
                    "MacAddress": "02:42:ac:11:00:02",
                    "NetworkID": "3c22b4b58101914cad4e7ab53cf301e4e4499bfc14e7bf47f2bf8ab058333b14",
                    "EndpointID": "730350d5da4b961daefb500c7803e4a37565a1a56cb7408a570fea5e34f822aa6",
                    "Gateway": "172.17.0.1",
                    "IPAddress": "172.17.0.2",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DriverOpts": null,
                    "DNSNames": null
                }
            }
        }
    }
]
```

10. Docker ps -a: Shows all containers

```
C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS                      PORTS
  NAMES
e91d072229ff   ubuntu    "/bin/bash"              27 minutes ago   Exited (1) 9 seconds ago
  silly_bhaskara
66841edd1628   ubuntu    "/bin/bash"              27 hours ago     Exited (0) 27 hours ago
  epic_swirles
833cbaa46af1   nginx     "/docker-entrypoint.…"   27 hours ago     Created
  laughing_hopper
7e1b86f9d578   nginx     "/docker-entrypoint.…"   27 hours ago     Created
  zealous_wiles
3a6123d69df1   ubuntu    "/bin/bash"              27 hours ago     Exited (0) 27 hours ago
  condescending_chaum
```

11. Docker stats: Display real-time resource usage statistics of running containers.

```
C:\Users\15L>docker stats
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
f1d175be2472   mycontainer   0.00%     1.996MiB / 15.49GiB   0.01%     1.02kB / 0B   0B / 0B     1
^C
```

12. Docker rm: Removes container

```
C:\Users\15L>docker rm -f mycontainer
mycontainer
```

13.Run nginx in the background

```
^C
C:\Users\15L>docker run -d --name web nginx
da94f70bca1900db7fbfd230ae926672cb0be1d839d0917b2b331d3ae395cc48

C:\Users\15L>
```

Linux Commands: Used to perform operations and manage files, directories, and processes inside a Linux system.

```
/
root@e91d072229ff:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@e91d072229ff:/# mkdir test
root@e91d072229ff:/# cd test
root@e91d072229ff:/test# touch file.txt
root@e91d072229ff:/test# cat file.txt
root@e91d072229ff:/test# echo "Hello Ubuntu" > file1.txt
root@e91d072229ff:/test# cat file1.txt
Hello Ubuntu
root@e91d072229ff:/test# cp file1.txt copy.txt
root@e91d072229ff:/test# mv copy.txt newfile.txt
root@e91d072229ff:/test# rm newfile.txt
root@e91d072229ff:/test# rmdir test
rmdir: failed to remove 'test': No such file or directory
root@e91d072229ff:/test# cd ..
root@e91d072229ff:/# rmdir test
rmdir: failed to remove 'test': Directory not empty
root@e91d072229ff:/# exit
exit
```

14. Docker Pull Alpine: Used to download the Alpine Linux image from Docker Hub.

```
C:\Users\15L>docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
55afa1ecc21d: Pull complete
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview alpine
```

15. Search for Image: Used to search and find available Docker images from Docker Hub using the docker search command.

```
C:\Users\15L>docker search python
NAME                      DESCRIPTION                                     STARS     OFFICIAL
python                    Python is an interpreted, interactive, objec…   10493     [OK]
circleci/python           Python is an interpreted, interactive, objec…   116
cimg/python                                                               25
demisto/python            The Demisto base python image                   5
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   33
okteto/python                                                             0
paketobuildpacks/python                                                   0
cleanstart/python         Deploy, scale, and secure critical infrastru…   1
activestate/python        ActiveState's customizable, low-to-no vulner…   11
intel/python                                                              5
corpusops/python          https://github.com/corpusops/docker-images/     1
rootpublic/python                                                         0
gentkit/python            A Docker images of Python environment .         0
stagex/python             High-level scripting language                   0
linuxserver/python                                                        0
vulhub/python                                                             2
koxudaxi/python           Dockerfiles for CPython of lysnikolaou's tag…   0
clarinpl/python                                                           0
unidata/python            Container with Unidata's python packages as …   3
dtcenter/python           Necessary Python3 packages for plotting GRIB…   1
openeuler/python                                                          1
dragonflyoss/python                                                       0
nanozoo/python                                                            0
hkube/python                                                              0
python/docker-bpo         Docker image for local development of http:/…   0

C:\Users\15L>
```

16. Docker Image Prune: Used to remove unused Docker images and free up storage space.

```
C:\Users\15L>docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
```

17. Docker Container LS: Used to list all currently running Docker containers.

```
C:\Users\15L>docker container ls -a
CONTAINER ID   IMAGE                              COMMAND                  CREATED         STATUS                          PORTS                    NAMES
3f9eed2969de   ubuntu                             "/bin/bash"              26 minutes ago  Exited (127) 19 minutes ago                              modest_perlman
16eccf1a92e9   ubuntu                             "/bin/bash"              30 minutes ago  Exited (0) 30 minutes ago                                kind_diffie
e8841e749448   sample-flask-app:v1                "flask run --host=0.…"   4 months ago    Created                                                  stupefied_brattain
a169e1386175   sample-flask-app:v1                "flask run --host=0.…"   4 months ago    Exited (255) 3 months ago       0.0.0.0:5000->5000/tcp   funny_newton
8e37dff4d72c   sample-flask-app:v1                "."                      4 months ago    Created                         0.0.0.0:5000->5000/tcp   angry_feistel
41ceff137f19   nginx                              "/docker-entrypoint.…"   4 months ago    Exited (255) 3 months ago       0.0.0.0:3000->80/tcp     quizzical_saha
05652b53729f   httpd                              "httpd-foreground"       4 months ago    Exited (255) 3 months ago       0.0.0.0:9090->80/tcp     focused_cerf
171bbef15299   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                                  objective_nobel
68e8ba592c7f   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                                  trusting_feistel
48f8e7a8ca1b   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                                  fervent_jepsen
3872caa94e58   nginx                              "/docker-entrypoint.…"   4 months ago    Created                                                  practical_banzai
341cff10e221   ca2b0f26964c                       "bash"                   4 months ago    Exited (127) 4 months ago                                hungry_napier
3da66845e08b   hello-world                        "/hello"                 4 months ago    Exited (0) 4 months ago                                  magical_heyrovsky
4f85ce4bf575   httpd                              "httpd-foreground"       2 years ago     Exited (255) 4 months ago       80/tcp                   determined_bohr
b34f2765cc66   ca2b0f26964c                       "/bin/bash"              2 years ago     Exited (0) 2 years ago                                   reverent_euclid
72f2f13072f3   welcome-to-docker:latest           "docker-entrypoint.s…"   2 years ago     Exited (255) 2 years ago        0.0.0.0:8089->3000/tcp   competent_hawking
b30cf9e8fa7e   welcome-to-docker:latest           "docker-entrypoint.s…"   2 years ago     Created                                                  zen_chaplygin
8dd6df41521d   docker/welcome-to-docker:latest    "/docker-entrypoint.…"   2 years ago     Exited (255) 2 years ago        0.0.0.0:8088->80/tcp     welcome-to-docker
```

18. Docker Info: Used to display detailed information about the Docker system and configuration.

```
C:\Users\15L>docker info
Client:
 Version:    25.0.3
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.12.1-desktop.4
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v2.24.6-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container. (Docker Inc.)
    Version:  0.0.24
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  dev: Docker Dev Environments (Docker Inc.)
    Version:  v0.1.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-dev.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.22
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  feedback: Provide feedback, right in your terminal! (Docker Inc.)
    Version:  v1.0.4
    Path:     C:\Program Files\Docker\cli-plugins\docker-feedback.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.0.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
    Version:  0.6.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-sbom.exe
  scout: Docker Scout (Docker Inc.)
    Version:  v1.5.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-scout.exe

Server:
 Containers: 6
  Running: 0
  Paused: 0
  Stopped: 6
 Images: 5
 Server Version: 25.0.3
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
```

19. Docker History: Used to show the history and layers of a Docker image.

```
C:\Users\15L>docker history ubuntu
IMAGE          CREATED      CREATED BY                                      SIZE      COMMENT
de7345b16e94   8 days ago   umoci raw add-layer --image /home/buildd/roc…   179B      Add rock control
 metadata
<missing>      8 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set annotations
<missing>      8 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set labels
<missing>      8 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set default PATH
 for bare-based rock
<missing>      8 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set default comm
ands
<missing>      8 days ago   umoci config --image /home/buildd/rockcraft-…   0B        Set entrypoint
<missing>      8 days ago   umoci raw add-layer --image /home/buildd/roc…   100MB

C:\Users\15L>
```

20. Disk Usage (docker system df): Used to display the amount of disk space used by Docker images, containers, and volumes.

```
C:\Users\15L>docker system df
TYPE            TOTAL     ACTIVE    SIZE      RECLAIMABLE
Images          10        1         1.523GB   1.362GB (89%)
Containers      1         1         1.095kB   0B (0%)
Local Volumes   1         0         0B        0B
Build Cache     16        0         0B        0B

C:\Users\15L>
```

21. Docker Network Inspect: Used to view detailed information about a Docker network.

```
C:\Users\15L>docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "3c22b4b58101914cad4e7ab53cf301e4e4499bfc14e7bf47f2bf8ab058333b14",
        "Created": "2026-07-20T09:49:36.630387228Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.17.0.0/16",
                    "Gateway": "172.17.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {
            "com.docker.network.bridge.default_bridge": "true",
            "com.docker.network.bridge.enable_icc": "true",
            "com.docker.network.bridge.enable_ip_masquerade": "true",
            "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
            "com.docker.network.bridge.name": "docker0",
            "com.docker.network.driver.mtu": "1500"
        },
        "Labels": {}
    }
]
```

22. Docker Volume: Used to create and manage persistent storage for Docker containers.

```
C:\Users\15L>docker volume create myvolume
myvolume
```

23. Docker Network: Used to create and manage communication between Docker containers.

```
C:\Users\15L>docker network create mynetwork
d7199082ae1108db3a65c9f47209db15831e3c38c3a8e8c67000693f4e0c8d52
```

24. Docker System Prune: Used to remove unused containers, images, networks, and cache to free Docker storage space.

```
C:\Users\15L>docker system prune
WARNING! This will remove:
  - all stopped containers
  - all networks not used by at least one container
  - all dangling images
  - unused build cache

Are you sure you want to continue? [y/N] Y
Deleted Containers:
e91d072229ffe0fccc81d980aa945db74397d3a4e0d600416b9ecbea139744f6
66841edd1628f268386be4a7ee7b4990beb2b03250e32bbbc9348392cafba7c2
833cbaa46af105b64eb1d177fc5003f11d51743284a5965905f69855dca3951b
7e1b86f9d5786209df527ad412703358f6af672c2f7ba5029f14c65aa1b25013
3a6123d69df18af58ff0bdbe6abe0b01cf411cd5171f1dbb6594e3611aa7d782

Total reclaimed space: 237B

C:\Users\15L>
```

25. Docker rename

```
C:\Users\15L>docker rename b125f8a6a583 ubuntu-test

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                      PORTS
    NAMES
b125f8a6a583   ubuntu         "/bin/bash"              33 minutes ago   Exited (0) 33 minutes ago
    ubuntu-test
5fde9bfdd162   ca2b0f26964c   "bash -c 'echo Hello…"   4 months ago     Exited (0) 4 months ago
    frosty_kepler
6587c9c4a2bf   nginx          "/docker-entrypoint.…"   4 months ago     Exited (0) 4 months ago
    charming_franklin
adefa716cb32   hello-world    "/hello"                 4 months ago     Exited (0) 4 months ago
    jolly_swanson
77ea50afe062   ca2b0f26964c   "bash -c 'echo Hello…"   4 months ago     Exited (0) 4 months ago
    agitated_agnesi
56c98c2698d7   ca2b0f26964c   "bash"                   4 months ago     Exited (0) 4 months ago
    pedantic_golick
```

Conclusion:

In this experiment, we understood how Docker helps in deploying and running applications without the need for manual installation of software and dependencies. We learned that Docker containers provide a lightweight and isolated environment, allowing applications to run smoothly and consistently across different systems. We also explored the concept of containerisation and understood the difference between Docker images and containers.